

Diabetic Prediction using Optimized Catboost Model

P. Preeti Payal, Research Scholar, Department of Computer Science Engineering (IT), University College of Engineering (A), Osmania University, Hyderabad. rpreeti.payal@gmail.com

Prof. P. Vijayapal Reddy, Professor and Head of the Department, Department of Computer Science Engineering, Matrusri college of Engineering, Saidabad. drpvijayapalreddy@gmail.com

Abstract

Diabetes is a major worldwide health concern, and successful management and prevention depend on early detection. The primary objective of this research is to enhance diabetes prediction accuracy using advanced machine learning techniques, specifically CatBoost, to address existing gaps in model interpretability and generalizability across diverse populations. A novel methodology was employed, utilizing the CatBoost algorithm within the gradient boosting framework. CatBoost is known for its superior handling of categorical data and robustness even with smaller datasets. Key innovations include target-based statistics, order-based encoding, and ordered boosting. Additionally, GridSearchCV was used for hyperparameter tuning to optimize the model's performance. The research involved the analysis of a comprehensive dataset from Kaggle, including various medical predictors such as age, BMI, insulin levels, and more. The results demonstrated that the CatBoost classifier achieved the highest accuracy of 92.8%, outperforming other classifiers like RandomForest and XGBoost.

Keywords: Diabetes Prediction, Machine Learning, CatBoost, Gradient Boosting, Hyperparameter Tuning, Healthcare Data.

1. Introduction

Using a variety of statistical and machine learning techniques, diabetic prediction based on data is carried out by evaluating patient data and estimating the risk of getting diabetes [1]. In order to build prediction models, this method uses historical data, which includes elements like age, blood pressure, level of glucose, body mass index (BMI), family medical history, along with lifestyle decisions. By calculating individual risk profiles, these models help with diabetes management and early identification [2]. Patterns and correlations within the data are analyzed to forecast the onset of diabetes, enabling healthcare professionals to implement preventive measures or early interventions. The significance of data-driven diabetic prediction is immense, given the global prevalence of diabetes, which affects millions and imposes a significant burden on healthcare systems [3]. Through early prediction, effective management and prevention of diabetes can be facilitated, leading to timely interventions that may delay or prevent the onset of the disease. Patients benefit from improved quality of life, reduced medical expenses, and a lower risk of complications such as cardiovascular disease, renal failure, and neuropathy [4]. Additionally, healthcare providers and policymakers are equipped with a strategic tool to efficiently allocate resources and design targeted public health initiatives. Various techniques are utilized in the prediction of diabetes using data. Traditional statistical methods, like logistic regression and decision trees, are favored for their simplicity and ease of interpretation. More advanced machine learning methods [5], such as neural networks, random forests (RF), and support vector machines (SVM), have been popular recently because of their higher accuracy and resilience and capacity to handle large and complicated datasets. The ability to mix the advantages of numerous models using ensemble learning approaches is also showing promise in improving prediction performance. Furthermore, the ability of deep learning techniques [6], such as recurrent neural networks (RNNs) and convolutional neural networks (CNNs), to recognize complex patterns in high-dimensional data is being researched.

Despite significant progress, several research gaps exist in the current methods for diabetic prediction [7]. One major gap is the challenge of interpretability in complex machine learning models, making it difficult for healthcare providers to comprehend and rely on the predictions. Moreover, most models

are developed based on data from specific populations, which may not be generalizable to diverse demographic groups, underscoring the need for more inclusive datasets and models that consider varying risk factors across different populations [8]. The integration of dynamic and real-time data, such as continuous glucose monitoring and data from wearable devices, into predictive models remains underexplored and could enhance the timeliness and personalization of predictions. Additionally, better validation of predictive models in real-world clinical settings is necessary to ensure their practicality and effectiveness in routine healthcare.

2. Literature

Varun Jaiswal et al [9] provided an overview of most studies on the use of machine learning and data mining techniques to forecast diabetes and its consequences. This research will help to improve the understanding of the diabetes pattern and the ability to forecast the illness. Diabetes is a metabolic disease characterized by persistently elevated blood glucose levels brought on by the body's improper use of glucose. Severe consequences such as diabetic ketoacidosis, nonketotic hyperosmolar coma, stroke, heart attack, chronic kidney failure, retinal impairment, and foot ulcers can result from this syndrome.

Isfazzaman Tasin et al [10] employed a private dataset of Bangladeshi female patients to build an autonomous diabetes prediction model with a variety of machine learning techniques. To supplement the utilization of the Pima Indian diabetes dataset, 203 additional samples were collected from a nearby textile industry in Bangladesh. In this work, the mutual information feature selection approach was used. The insulin properties in the private dataset were predicted using a semi-supervised model with significant gradient boosting. To address class imbalance, SMOTE and ADASYN techniques were implemented.

Jobeda Jamal Khanam et al [11] used the UCI Machine Learning Repository's Pima Indian Diabetes (PID) dataset for this study. There are nine different features in this dataset, which includes information on 768 patients. This dataset was subjected to seven machine learning algorithms in order to predict diabetes. It was discovered that the models using SVM and Logistic Regression (LR) were successful in predicting diabetes.

Neha Prerna Tigga et al [12] evaluated the risk of diabetes in individuals by considering their lifestyle and family history. Various machine learning algorithms were employed to predict the risk of Type 2 diabetes due to their high accuracy, which is essential in the healthcare field. Once these models achieve a high level of accuracy, individuals will be able to self-assess their diabetes risk.

Md. Kamrul Hasan et al [13] provided a robust framework for predicting diabetes, utilizing Multilayer Perceptrons (MLP), K-fold cross-validation, data standardization, feature selection, and outlier rejection. There were several Machine Learning (ML) classifiers used. To increase the accuracy of diabetes prediction, the study also suggested using weighted ensembling of many machine learning models. The Area Under the ROC Curve (AUC) of each model was used to determine the ensemble weights. During the hyperparameter tuning procedure, the grid search approach was employed to maximize the major performance statistic, AUC.

Md. Maniruzzaman et al [14] created a machine learning (ML) algorithm to predict if patients have diabetes. Based on the p-value along with odds ratio (OR), logistic regression (LR) was utilized to determine the risk variables for diabetic sickness. To forecast the diabetic patients, the authors have used four ML classifiers. Additionally, these protocols have been duplicated and incorporated into 20 trails using three other types of partition procedures (K2, K5, and K10).

Jayroop Ramesh et al [15] suggested that smartphones, smart wearables, and personal health devices be utilized to establish an automated framework for comprehensive remote monitoring aimed at predicting and managing diabetes risk. After feature scaling, imputation, selection, as well as augmentation procedures, a support vector machine was created for diabetes risk prediction employing the Pima Indian Diabetes Database.

Chollette C. Olisah et al [16] suggested developing a strong framework to help with the clinical diagnosis of diabetes by developing a diabetes prediction model. The system is designed to integrate polynomial regression for feature selection and Spearman correlation for imputing missing values, enhancing their respective performances. For classification problems, other supervised machine learning models are also offered, such as the SVM model, the RF model, and a specially-designed twice-growth deep neural network (2GDNN). These models are evaluated for their scalability to the prediction problem after undergoing optimization through hyperparameter tuning utilized grid search along with repeated stratified k-fold cross-validation.

3. Proposed Method

Accurately predicting when diabetes may strike is essential for prompt action and efficient treatment of the condition. This study presents a unique method for diabetes prediction that makes use of the potent machine learning algorithm CatBoost. CatBoost is a powerful tool for managing categorical data that continues to perform well with smaller datasets. We want to find and examine intricate patterns and linkages in medical datasets that indicate a higher risk of diabetes by utilizing CatBoost's advanced capabilities.

3.1 CatBoost

The gradient boosting algorithm known as "Categorical Boosting," or "CatBoost," to handle category data more efficiently and outperform more conventional gradient boosting techniques in terms of speed and accuracy. An extensive explanation of the main parts and constituents of the CatBoost architecture can be found below.

A. Gradient Boosting Framework

The gradient boosting framework is where CatBoost operates at its core. A machine learning technique called gradient boosting builds a prediction model for regression as well as classification challenges piecemeal. It entails gradually introducing weak learners (often decision trees) to the ensemble, each of which fixes the mistakes of the one before it. This approach is effective since it can continuously enhance the model's performance.

B. Handling of Categorical Features

CatBoost's ability to handle categorical data well is one of its best qualities. Traditional gradient boosting algorithms sometimes struggle with categorical data, needing them to be translated into numerical form by one-hot encoding or other preprocessing procedures. Nonetheless, CatBoost uses the following techniques to handle categorical features:

- **Target-Based Statistics:** CatBoost uses statistics to convert categorical information into numerical values, such as the mean goal value, or it can apply additional specific transformations based on the features. This method preserves the categorical context within the data by capturing the connection between the target variable and the category feature.
- **Order-Based Encoding:** Different categorical characteristics are encoded during training and inference by CatBoost in order to prevent overfitting and guarantee the robustness of the model. It uses a method known as "order-based encoding," in which the order of the samples determines the transformation. This successfully prevents information from future samples from leaking out since each data point is encoded depending on the data that has already been viewed.

C. Ordered Boosting

"Ordered boosting," a revolutionary method of building decision trees, is introduced by CatBoost. Overfitting can occur when traditional gradient boosting techniques utilize the same data for both tree construction and learning. Ordered boosting counteracts this by employing a permutation-based strategy that keeps each sample "online" by only utilizing it once for training. This technique reduces overfitting and improves the model's capacity for generalization by dividing the training set into

numerous parts. Each part is then used to create trees, and the predictions are based on trees trained exclusively on the previous parts.

D. Combination of Multiple Loss Functions

CatBoost can handle a wide range of tasks, such as regression, classification, and ranking, since it supports a large number of loss functions. By optimizing the selected loss function during training, it offers unique solutions for particular kinds of issues. Several of the often employed loss functions in CatBoost include of:

- **Logloss:** Utilized in tasks involving binary classification.
- **RMSE:** Employed to reduce the root mean squared error in regression tasks..
- **Cross-Entropy:** Applied in classification tasks involving multiple classes.
- **QuerySoftMax:** Utilized to maximize the ranking metrics in ranking tasks.

E. Efficient Training and Prediction

CatBoost is made to function well during both the training and prediction stages. It uses the power of the CPU and GPU to speed up processing. Important methods employed are as follows:

- **Gradient Bias Correction:** By adjusting the gradients, CatBoost uses a bias correction approach to overcome the problem of bias in gradient estimation, resulting in training that is more accurate and stable.
- **Symmetric Trees:** When CatBoost creates symmetric trees, every branch breaks on the same feature at every level of the tree. When compared to other gradient boosting techniques that create asymmetric trees, CatBoost is more effective because of its symmetry, which dramatically accelerates the training and prediction stages.

F. Regularization and Overfitting Prevention

To further combat overfitting, CatBoost includes several regularization techniques:

- **L2 Regularization:** Adds a penalty term proportional to the sum of the squared coefficients, discouraging the model from becoming too complex.
- **Shrinkage (Learning Rate):** This technique scales the contribution of each weak learner, reducing the model's sensitivity to the added weak learners and thus smoothing the final predictions.
- **Depth Control:** By controlling the maximum depth of the trees, CatBoost limits the model complexity and prevents it from fitting the noise in the data.

G. Cross-Validation and Model Selection

Strong tools for model selection and cross-validation are provided by CatBoost. It supports time-series split and classic k-fold cross-validation, which are crucial for evaluating model performance and making ethical hyperparameter selections. This guarantees that the selected model is well-tuned and has good generalization to new data..

The architecture of CatBoost is designed to overcome the difficulties in managing categorical data and enhancing the effectiveness and precision of gradient boosting. Through the integration of novel techniques such as ordered boosting, target-based statistics for categorical features, and effective training mechanisms, CatBoost offers a robust and adaptable instrument for various machine learning assignments. Its versatility and capacity to avert overfitting make it appropriate for use in a wide range of industries, including healthcare and banking.

CatBoost is particularly good at handling categorical data natively, which simplifies data preparation by removing the requirement for one-hot encoding and other preprocessing. By preventing data leaking during training, its novel ordered boosting strategy lowers overfitting and improves model generalization. The technique can quickly train and provide predictions on big datasets without compromising accuracy because to its effective use of GPU acceleration and symmetric tree building. Thanks to these characteristics, CatBoost performs exceptionally well on a wide range of machine learning tasks while being very intuitive and effective.

Because of its adaptability, CatBoost is perfect for a wide range of applications in many sectors. It is used in finance for jobs where accuracy and management of complicated data are essential, such as stock market prediction, fraud detection, and credit scoring. CatBoost uses its capacity to efficiently store categorical patient data to assist in illness diagnosis, patient outcome prediction, and customized treatment planning in the healthcare industry. Because of its quick processing speed and strong performance, e-commerce platforms employ it for recommendation systems, revenue forecasting, and user churn prediction. Its versatility and strength across a range of industries are further demonstrated by its use in manufacturing and telecommunications for anomaly identification and improvements in operational efficiency.

Algorithm Steps for CatBoost

Step 1: Initialize Parameters: Set the number of iterations and Initialize model parameters.

Step 2: Data Preparation: Handle missing values and Encode categorical features.

Step 3: Initialize Model: Create an initial prediction, often set to zero or a constant.

Step 4: Gradient Computation: Compute gradients based on the current model prediction.

Step 5: Weighted Average: Compute a weighted average of the gradients for each category of categorical features.

Step 6: Tree Construction: Build a decision tree using the weighted gradients.

Step 7: Update Model: Update the model prediction using the new tree.

Step 8: Shrinkage: Apply a learning rate to scale the updates.

Step 9: Repeat: Repeat steps 4 to 8 for the specified number of iterations.

Step 10: Final Model: Aggregate the trees to form the final prediction.

Step 10: Model Evaluation: Evaluate the model using a specified metric.

3.2 Hyperparameter tuning

Hyperparameters are configuration options that have an impact on the training process and model performance, making them essential parts of machine learning and statistical modeling. Hyperparameters control how the learning process proceeds and are specified before the learning process starts, in contrast to model parameters, which are learnt immediately from the training data. The learning rate in neural networks, the number of clusters in k-means clustering, the regularization parameter in regression models, and the depth of trees in decision trees are a few examples of hyperparameters. The model's accuracy, capacity for generalization, and rate of convergence may all be strongly impacted by the selection of hyperparameters. Hyperparameters are generally selected by trial and error, gut feeling, or hyperparameter tuning techniques in order to attain the optimal model performance; they are not optimized by the training algorithms.

The process of determining the ideal collection of hyperparameters for a particular model is called hyperparameter tuning, or hyperparameter optimization. The correct selection of hyperparameters may increase generalization to new data, decrease overfitting, and improve model performance, which makes this approach essential. Hyperparameter tuning may be done in a number of ways, such as grid search, random search, and more complex techniques like gradient-based optimization and Bayesian optimization. Random search takes random samples of hyperparameters from a predetermined distribution, whereas grid search searches across a manually selected subset of the hyperparameter space in an exhaustive manner. More sophisticated techniques, like Bayesian optimization, make use of probabilistic models to assess how well certain hyperparameter settings work. They then repeatedly narrow down the search space to concentrate on the most promising regions. Although it might be time-consuming and computationally demanding, hyperparameter tweaking is crucial for creating reliable and accurate models.

3.2.1 GridSearchCV

A crucial tool in the machine learning toolkit for maximizing model performance through hyperparameter adjustment is GridSearchCV. Hyperparameters are predefined settings of a model that

cannot be derived from the data and need to be determined before the training process begins. For example, the depth of a decision tree, which influences its complexity, and the learning rate, which controls the speed at which the model learns, are key hyperparameters.

A CatBoost Classifier, a kind of gradient boosting model well-known for its effectiveness with categorical data, is refined using GridSearchCV. The first step in the procedure is to define a collection of hyperparameters to investigate. In this case, the hyperparameters ``learning_rate`` and ``depth`` are mentioned. The amount by which the model's weights are modified during each learning iteration is determined by the `{learning_rate}`; lesser values generally result in more gradual and maybe more robust learning. The maximum depth of the decision trees employed in the boosting process is set by the ``depth`` parameter, which affects the model's capacity to identify intricate patterns in the data.

GridSearchCV does a grid search to methodically identify the ideal set of these hyperparameters. This indicates that all feasible combinations of the given hyperparameters are evaluated. In this instance, it will experiment with several settings of ``depth`` (3, 5, 7, 9), and ``learning_rate`` (0.01, 0.05, 0.1). Every combination stands for a different set of testable model configurations.

Cross-validation, often abbreviated as "CV," is a technique used in GridSearchCV to assess how well a model performs on unseen data. This method involves splitting the dataset into multiple sections, called "folds." The model is then repeatedly trained on some of these folds while the remaining fold is used to validate the model's performance. Five-fold cross-validation (`cv=5`) was applied in this instance, dividing the data into five sections. Five training cycles are applied to the model, with a distinct portion awaiting validation at each iteration. This procedure lessens the unpredictability that may result from utilizing a single train-test split, which aids in getting a trustworthy assessment of the model's performance.

When initialized, ``GridSearchCV`` combines these elements: it will test all specified combinations of ``learning_rate`` and ``depth`` using 5-fold cross-validation. This ensures that each set of hyperparameters is evaluated rigorously, providing a thorough search for the best model configuration. For the CatBoost Classifier used in this example, the ``verbose=False`` parameter suppresses detailed output during training, making the tuning process more streamlined and less noisy.

Finding a model's ideal hyperparameters through trial and error is automated using GridSearchCV. The objective is to determine the optimal parameters for the CatBoost Classifier by thoroughly examining all possible combinations of `{learning_rate}` and `{depth}`, and then verifying each combination with 5-fold cross-validation. This technique avoids overfitting and guarantees that the selected model generalizes well to new, unknown data, in addition to improving model accuracy.

To improve the accuracy of diabetes onset predictions, this study presents an innovative technique using the CatBoost machine learning algorithm. Our approach harnesses CatBoost's sophisticated features within the gradient boosting framework to develop a robust predictive model. Additionally, we implement efficient training methods and employ GridSearchCV for hyperparameter tuning to fine-tune model performance. By combining these elements, our method aspires to provide precise and dependable predictions for diabetes, facilitating timely and effective management.

4. Experimental Results

A thorough examination of the outcomes of the simulations carried out with the suggested approach is provided in this section. The Pima Indian Diabetes Dataset is a prominent benchmark in machine learning and healthcare analytics. It was developed from research conducted by the National Institute of Diabetes and Digestive and Kidney Diseases. This dataset contains medical and demographic data from a group of Pima Indian women. It includes 768 records, each with eight attributes: the number of pregnancies, glucose levels, blood pressure, skinfold thickness, insulin levels, body mass index (BMI), diabetes pedigree function, and age. The goal is to classify whether each individual has diabetes based on these attributes. This dataset is particularly valuable due to its real-world relevance and the challenge it poses for predictive modeling; the data includes non-linear and complex relationships

between the features and the target variable. The patient's age, BMI, insulin level, number of pregnancies, and other factors are examples of predictor variables.

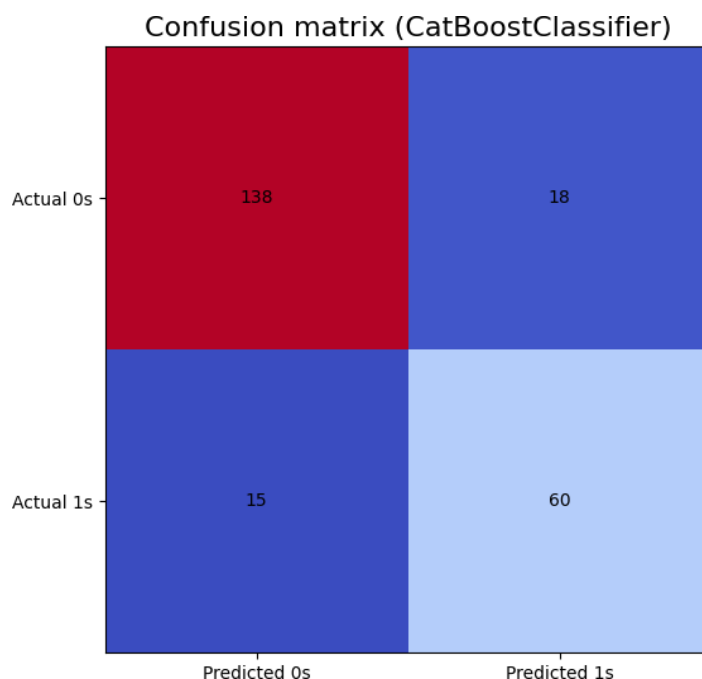


Figure 1: Confusion Matrix

The confusion matrix shown in Figure 1 provides a comprehensive overview of how well the CatBoostClassifier predicts whether individuals have a condition like diabetes. This 2x2 matrix compares actual outcomes (true labels) with the predicted outcomes produced by the model. Each cell in the matrix represents a specific scenario of true or false predictions, giving insights into the model's accuracy and highlighting areas where errors occur.

- **True Negatives (TN) - Top Left Cell (138):**

This cell indicates the number of cases where the model accurately predicted the absence of diabetes. Here, 138 instances were correctly classified as not having diabetes (actual class 0, predicted class 0). This demonstrates the model's effectiveness in identifying individuals without the condition.

- **False Positives (FP) - Top Right Cell (18):**

The top right cell shows instances where the model incorrectly predicted the presence of diabetes. In 18 cases, individuals were wrongly identified as having diabetes (actual class 0, predicted class 1). These are Type I errors or false alarms, where the model overestimates the occurrence of diabetes.

- **False Negatives (FN) - Bottom Left Cell (15):**

The bottom left cell displays the number of instances where the model failed to identify diabetes. There were 15 cases where individuals had diabetes, but the model incorrectly predicted they did not (actual class 1, predicted class 0). These are Type II errors or misses, indicating the model's tendency to overlook actual cases of diabetes.

- **True Positives (TP) - Bottom Right Cell (60):**

This cell represents the number of times the model correctly identified the presence of diabetes. Sixty instances were accurately predicted as having diabetes (actual class 1, predicted class 1). This reflects the model's capability to correctly detect individuals with the condition.

Table 1: Comparative Analysis

Method	Accuracy
SGDClassifier	0.736
LogisticRegression	0.783
SVC	0.861
DecisionTreeClassifier	0.877

LGBMClassifier	0.911
KNeighborsClassifier	0.914
XGBClassifier	0.915
RandomForestClassifier	0.918
CatBoostClassifier	0.928

Table 1 provides a comparative overview of the accuracy of different machine learning classifiers. The CatBoostClassifier stands out with the highest accuracy at 0.928, followed closely by the RandomForestClassifier at 0.918 and the XGBClassifier at 0.915. Both KNeighborsClassifier and LGBMClassifier also exhibit strong performance, with accuracies of 0.914 and 0.911, respectively. In contrast, the DecisionTreeClassifier, SVC, and LogisticRegression show moderate accuracy, ranging from 0.877 to 0.783. The SGDClassifier, with an accuracy of 0.736, performs the least effectively. This analysis underscores the CatBoostClassifier's leading performance in this context.

5. Conclusion

The research demonstrated a significant advancement in diabetes prediction using the CatBoost algorithm, achieving an impressive accuracy of 92.8%, the highest among all tested classifiers. This model effectively reduced false positives and false negatives, as shown by the confusion matrix, underscoring its robustness and reliability in identifying diabetes onset. These results highlight the model's potential to serve as a powerful tool in early diabetes detection and personalized healthcare management. The implications of these findings are profound. By leveraging CatBoost's advanced handling of categorical data and its innovative approach to preventing overfitting, the study offers a scalable and interpretable solution to a critical healthcare challenge.

References

- [1] Bhat, Salliah Shafi, Venkatesan Selvam, Gufran Ahmad Ansari, Mohd Dilshad Ansari, and Md Habibur Rahman. "Prevalence and early prediction of diabetes using machine learning in North Kashmir: a case study of district bandipora." *Computational Intelligence and Neuroscience* 2022, no. 1 (2022): 2789760.
- [2] Fregoso-Aparicio, Luis, Julieta Noguez, Luis Montesinos, and José A. García-García. "Machine learning and deep learning predictive models for type 2 diabetes: a systematic review." *Diabetology & metabolic syndrome* 13, no. 1 (2021): 148.
- [3] Ahmad, Hafiz Farooq, Hamid Mukhtar, Hesham Alaqail, Mohamed Seliaman, and Abdulaziz Alhumam. "Investigating health-related features and their impact on the prediction of diabetes using machine learning." *Applied Sciences* 11, no. 3 (2021): 1173.
- [4] Gupta, Himanshu, Hirdesh Varshney, Tarun Kumar Sharma, Nikhil Pachauri, and Om Prakash Verma. "Comparative performance analysis of quantum machine learning with deep learning for diabetes prediction." *Complex & Intelligent Systems* 8, no. 4 (2022): 3073-3087.
- [5] Joshi, Ram D., and Chandra K. Dhakal. "Predicting type 2 diabetes using logistic regression and machine learning approaches." *International journal of environmental research and public health* 18, no. 14 (2021): 7346.
- [6] Naz, Huma, and Sachin Ahuja. "Deep learning approach for diabetes prediction using PIMA Indian dataset." *Journal of Diabetes & Metabolic Disorders* 19 (2020): 391-403.
- [7] Khaleel, Fayroza Alaa, and Abbas M. Al-Bakry. "Diagnosis of diabetes using machine learning algorithms." *Materials Today: Proceedings* 80 (2023): 3200-3203.
- [8] Ahmed, Usama, Ghassan F. Issa, Muhammad Adnan Khan, Shabib Aftab, Muhammad Farhan Khan, Raed AT Said, Taher M. Ghazal, and Munir Ahmad. "Prediction of diabetes empowered with fused machine learning." *IEEE Access* 10 (2022): 8529-8538.
- [9] Jaiswal, Varun, Anjali Negi, and Tarun Pal. "A review on current advances in machine learning based diabetes prediction." *Primary Care Diabetes* 15, no. 3 (2021): 435-443.

- [10] Tasin, Isfuzzaman, Tansin Ullah Nabil, Sanjida Islam, and Riasat Khan. "Diabetes prediction using machine learning and explainable AI techniques." *Healthcare technology letters* 10, no. 1-2 (2023): 1-10.
- [11] Khanam, Jobeda Jamal, and Simon Y. Foo. "A comparison of machine learning algorithms for diabetes prediction." *Ict Express* 7, no. 4 (2021): 432-439.
- [12] Tigga, Neha Prerna, and Shruti Garg. "Prediction of type 2 diabetes using machine learning classification methods." *Procedia Computer Science* 167 (2020): 706-716.
- [13] Hasan, Md Kamrul, Md Ashraful Alam, Dola Das, Eklas Hossain, and Mahmudul Hasan. "Diabetes prediction using ensembling of different machine learning classifiers." *IEEE Access* 8 (2020): 76516-76531.
- [14] Maniruzzaman, Md, Md Jahanur Rahman, Benojir Ahammed, and Md Menhazul Abedin. "Classification and prediction of diabetes disease using machine learning paradigm." *Health information science and systems* 8 (2020): 1-14.
- [15] Ramesh, Jayroop, Raafat Aburukba, and Assim Sagahyroon. "A remote healthcare monitoring framework for diabetes prediction using machine learning." *Healthcare Technology Letters* 8, no. 3 (2021): 45-57.
- [16] Olisah, Chollette C., Lyndon Smith, and Melvyn Smith. "Diabetes mellitus prediction and diagnosis from a data preprocessing and machine learning perspective." *Computer Methods and Programs in Biomedicine* 220 (2022): 106773.